Solved selected problems of Information Theory, Inference, and Learning Algorithms by David J.C. MacKay.

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1 - Introduction to Information Theory

Solution. **Exercise 1.2**
When using R_3 we get an error if two or three bits are flipped in a set of 3 bits. Then the probability of two bits being flipped is

$$P(r = 2 \mid f, N = 3) = \binom{3}{2} f^2 (1-f)^{3-2} = 3f^2(1-f)$$

And the probability of three bits being flipped is

$$P(r = 3 \mid f, N = 3) = \binom{3}{3} f^3 (1-f)^{3-3} = f^3$$

Then the total probability of getting an error when using R_3 is

$$P(2 \mid f, 3) + P(3 \mid f, 3) = f^3 + 3f^2(1-f) = 3f^2 - 2f^3$$

We observe that $3f^2 - 2f^3$ is less than f if $f < 0.5$, hence, the error probability is reduced by the use of R_3. $\square$